YUNHUI JANG

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RESEARCH INTEREST

My research goal is to develop reasoning frameworks for AI-driven scientific discovery. Specifically, I focus on reasoning for science, generative models for structured data, and agentic systems for science. My previous research focused on:

- LLM Reasoning for science: C5, C6, C8, C9
- Generative models for structured data: C1, C2, C4, P1
- Agentic system for science: C6, C9, W1, W2

WORK EXPERIENCE

Genentech, Research intern, (Manager: Taylor Joren, Sidney Lisanza) San Francisco, USA / Jun 2026 - Aug 2026
• Conducted research on self-evolving memory construction for protein optimization
• MEMIG: Self-evolving Memory Construction for Antibody Optimization (NeurIPS 2026 AgenticLS Workshop)
Valence Labs, Research intern, (Manager: Emmanuel Noutahi) Montreal, Canada / Jul 2025 - Dec 2025
• Conducted research on constructing LLM agent for reasoning and developing reasoning models for virtual cells
• Towards Autonomous Mechanistic Reasoning in Virtual Cells (EMNLP 2026 Findings)
Netmarble, Research intern Seoul, South Korea / Jul 2018 - Aug 2018
• RNN-based anomaly detection
JoyCity, Intern Seoul, South Korea / Jun 2017 - Aug 2017
• Data analysis on marketing KPIs

EDUCATION

KAIST, Ph.D. student in Graduate School of AI (advisor: Prof. Sungsoo Ahn) Feb 2025 - Aug 2027 (expected)
POSTECH, M.S. in Graduate School of AI (advisor: Prof. Sungsoo Ahn) Sep 2022 - Feb 2025
RWTH, Exchange student in Computer Science Mar 2019 - Feb 2020
POSTECH, B.S. in Industrial & Management Engineering - cum laude Mar 2015 - Aug 2020

PUBLICATIONS

C: conference, J: journal, W: workshop, P: preprint / * equal contribution

[P1] Can LLMs Generate Diverse Molecules? Towards Alignment with Structural Diversity
Hyosoon Jang, Yunhui Jang, Jaehyung Kim, Sungsoo Ahn

[W2] MEMIG: Self-evolving Memory Construction for Antibody Optimization
Yunhui Jang, Zichen Wang, Jai Doshi, Kangway Chuang, Sidney Lisanza, Taylor Joren
NeurIPS AgenticLS Workshop, 2026

[W1] INDIBATOR: Diverse and Fact-Grounded Individuality for Multi-Agent Debate in Molecular Discovery
Yunhui Jang, Seonghyun Park, Jaehyung Kim, Sungsoo Ahn
ICML AI4Science Workshop, 2026

[C10] A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning
Hyosoon Jang, Hyunjin Seo, Honghui Kim, Taewon Kim, Yunhui Jang, Seonghyun Park, Sungsoo Ahn
Conference on Neural Information Processing Systems (NeurIPS), 2026

[C9] Towards Autonomous Mechanistic Reasoning in Virtual Cells
Yunhui Jang, Lu Zhu, Jake Fawkes, Alisandra Kaye Denton, Dominique Beaini, Emmanuel Noutahi
Conference on Empirical Methods in Natural Language Processing (EMNLP) (Findings), 2026
ICML GenBio Workshop, Spotlight, 2026

[C8] Improving Chemical Understanding of LLMs via SMILES Parsing
Yunhui Jang, Jaehyung Kim, Sungsoo Ahn
Conference on Empirical Methods in Natural Language Processing (EMNLP) (Main), 2025

[C7] Self-Training Large Language Models with Confident Reasoning
Hyosoon Jang, Yunhui Jang, Sungjae Lee, Jungseul Ok, Sungsoo Ahn
Conference on Empirical Methods in Natural Language Processing (EMNLP) (Findings), 2025

[C6] MT-MOL: Multi Agent System with Tool-based Reasoning for Molecular Optimization
Hyomin Kim, Yunhui Jang, Sungsoo Ahn
Conference on Empirical Methods in Natural Language Processing (EMNLP) (Findings), 2025

[C5] Structural Reasoning Improves Molecular Understanding of LLM
Yunhui Jang, Jaehyung Kim, Sungsoo Ahn
Annual Meeting of the Association for Computational Linguistics (ACL) (Main), 2025
NeurIPS AIDrugX Workshop, 2024

[C4] Pessimistic Backward Policy for GFlowNets
Hyosoon Jang, Yunhui Jang, Minsu Kim, Jinkyoo Park, Sungsoo Ahn
Conference on Neural Information Processing Systems (NeurIPS), 2024

[C3] Hybrid neural representations for spherical data
Hyomin Kim, Yunhui Jang, Jaeho Lee, Sungsoo Ahn
International Conference on Machine Learning (ICML), 2024

[C2] A simple and scalable representation for graph generation
Yunhui Jang, Seul Lee, Sungsoo Ahn
International Conference on Learning Representations (ICLR), 2024
NeurIPS GLFrontiers Workshop, 2023

[C1] Graph generation with K^2-trees
Yunhui Jang, Dongwoo Kim, Sungsoo Ahn
International Conference on Learning Representations (ICLR), 2024
ICML Structured Probabilistic Inference & Generative Modeling Workshop, 2023
Bronze Prize in 30th Samsung Humantech Paper Awards
Winner in 2024 Qualcomm Innovation Fellowship Korea

HONORS & AWARDS

Research Subsidies for Ph.D. Candidates, NRF ($40,000) 2025-present
Winner in Qualcomm Innovation Fellowship Korea, Qualcomm ($3,000) 2024
Presidential Science Scholarship, South Korea ($70,000) 2024-present
Travel Award, ICLR ($1,500) 2024
Bronze Prize in 30th Samsung Humantech Paper Awards, Samsung, ($4,000) 2023
POSTECHIAN Fellowship, POSTECH, ($3,000) 2022, 2023
RWTH Aachen University Exchange Scholarship, RWTH Aachen, ($3,900) 2019
Realize Your Dream Scholarship, Blizzard Entertainment, ($2,500) 2018
Presidential Science Scholarship, South Korea, ($16,000) 2015-2020

SERVICES

Area Chair: ARR 2026 May, ARR 2026 October
Conference Reviewer: NeurIPS 2023-2025, ICLR 2024-2027, ICML 2024-2026, AAAI 2025-2027
Workshop Reviewer: ICLR 2026 GEM, ICML 2026 AI4Science, ICML 2026 GFM, ICML 2026 GenBio, NeurIPS 2026 AI4Science, NeurIPS 2026 GEM
Workshop Organizer: ICML 2026 AI4Science

TALKS

Towards Autonomous Mechanistic Reasoning in Virtual Cells
GenBio AI, Palo Alto (Hosted by: Le Song) Jun 2026
Improving Structural Understanding of Molecular LLMs
AI Alliance, AI in Materials & Chemistry Webinar Series Aug 2025

LG AI research, Seoul Jun 2025

Graph Generation with K^2-trees

Qualcomm, Seoul Mar 2025

Speaking the Structure: Generative Models in Molecular Science

MILA, Montreal Dec 2024

TEACHING

Teaching Assistant, POSTECH, CSED105: Introduction to AI	Sep 2023 - Dec 2023
Teaching Assistant, LAIDD, Introduction to Geometric deep learning	Oct 2023 - Nov 2023
Teaching Assistant, POSTECH, CSED490B: Introduction to machine learning	Sep 2022 - Dec 2022
Teaching Assistant, Hyundai Steel, AI expert course	Oct 2022 - Nov 2022